English

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ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025-2026)

Test Pattern

NEET (UG)

MAJOR

30-03-2026

PRE-MEDICAL : LEADER & ACHIEVER COURSE PHASE - MLA,M,N,W,X,MAZC,D,E,O,P,U/V,MAAK,U/V,Y,Z,MAPB,MSP (SPARK PRO)

Test Booklet Code

L5

Booklet contains **28** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
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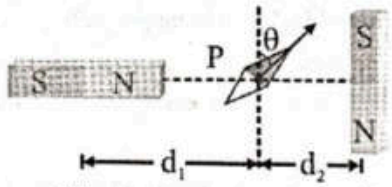
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SUBJECT : PHYSICS

Topic : SYLLABUS-5.

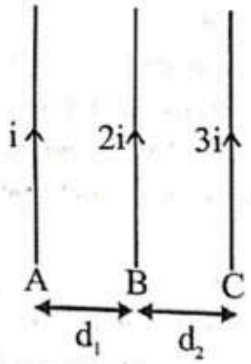
1. Two magnets A and B are identical and these are arranged as shown in the figure. Their length is negligible in comparison to the separation between them. A magnetic needle is placed between the magnets at point P which gets deflected through an angle θ under the influence of magnets. The ratio of distance d_1 and d_2 will be :-



(1) $(2\tan\theta)^{1/3}$ (2) $(2\tan\theta)^{-1/3}$

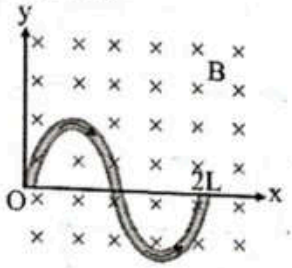
(3) $(2\cot\theta)^{1/3}$ (4) $(2\cot\theta)^{-1/3}$

2. In adjacent figure, all the wires are long and parallel, if net force per unit length on wire B is zero, then $\frac{d_1}{d_2}$ will be :-



(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{6}$ (4) $\frac{1}{4}$

3. A wire carrying a current i is placed in a uniform magnetic field is in the form of the curve $y = a\sin\left(\frac{\pi x}{L}\right)$, $0 \le x \le 2L$. The force acting on the wire is :-



(1) $\frac{iBL}{\pi}$ (2) $iBL\pi$ (3) 2iBL (4) Zero

4. A charged particle is projected in uniform magnetic field B with velocity v, having charge q perpendicular to the magnetic field. Then the force on charged particle is given by $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$. Comment about the angular frequency of charged particle :-

(1) remians constant for every charged particle

(2) increases with increase in radius of path

(3) increase with decrease in radius of path

(4) remains constant whatever may be velocity of projection of given charged particle

5. A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross-section. The ratio of the magnetic field at distance $\frac{a}{2}$ and 2a from axis of wire is-

(1) $\frac{1}{4}$

(2) 4

(3) 1

(4) $\frac{1}{2}$

6. The work done in rotating a magnet of magnetic moment M by an angle of 90° from the external magnetic field direction is 'n' times the corresponding work done to turn it from external magnetic field direction through an angle of 60°, where 'n' gives by :-

(1) 1/2

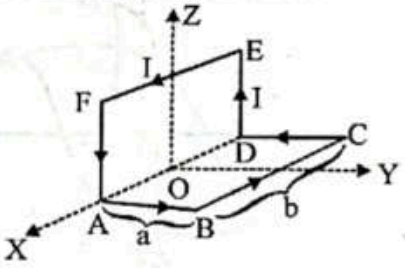
(2) 2

(3) 1/4

(4) 1

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7. A wire carrying current I is bent in the shape ABCDEFA as shown, where rectangle ABCDA and ADEFA are perpendicular to each other. If the sides of the rectangles are of lengths a and b, then the magnitude and direction of magnetic moment of the loop ABCDEFA is :



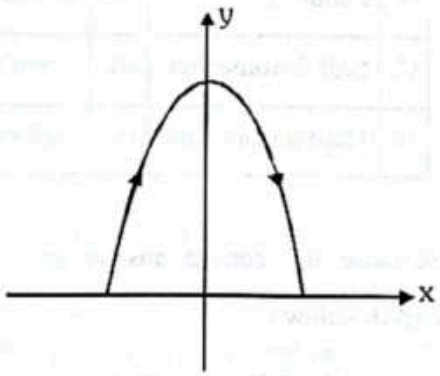
(1) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$

(2) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

(3) abI, along $\left(\frac{\hat{j}}{\sqrt{2}}+\frac{\hat{k}}{\sqrt{2}}\right)$

(4) abI, along $\left(\frac{\hat{j}}{\sqrt{5}}+\frac{2\hat{k}}{\sqrt{5}}\right)$

8. A wire carrying a current of 4A is bent in the form of a parabola $x^2 + y = 16$ as shown in figure, where x and y are in meter. The wire is placed in a uniform magnetic field $\vec{B} = 5\hat{k}$ tesla. The force acting on the wire is



(1) $80\,\hat{j}$ N

(2) $-80\,\hat{j}$ N

(3) $-160\,\hat{j}$ N

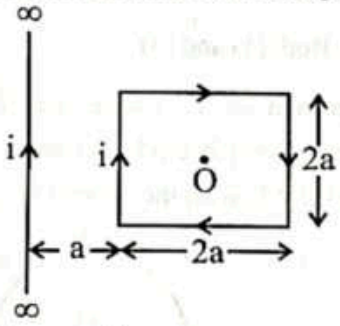
(4) $160\,\hat{j}$ N

9. Radius of the current carrying coil is R. If magnetic field at any point on the axis of the coil is $B_x = \frac{B_0}{64}$ then find out axial distance of this point :-
[where B_0 = magnetic field at center of coil]

(1) $\sqrt{3}R$ (2) $\sqrt{5}R$

(3) $\sqrt{15}R$ (4) R

10. Find net magnetic field at centre of square.



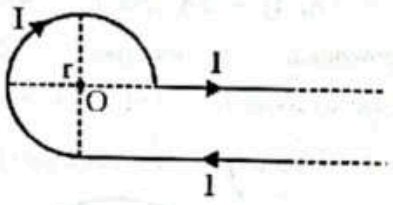
(1) $\frac{\mu_0 i}{2\pi a}\left[\frac{1}{2}+2\sqrt{2}\right]$

(2) $\frac{\mu_0 i}{2\pi a}\left[\frac{1}{2}+4\sqrt{2}\right]$

(3) $\frac{\mu_0 i}{2\pi a}\left[1+2\sqrt{2}\right]$

(4) $\frac{\mu_0 i}{2\pi a}\left[1-2\sqrt{2}\right]$

11. Current 'I' is flowing in a conductor shaped as shown in the figure. The radius of the curves part is r and the length of straight portion is very large. The value of the magnetic field at the centre O will be :-



(1) $\frac{\mu_0 I}{4\pi r}\left(\frac{\pi}{2}+1\right)$

(2) $\frac{\mu_0 I}{4\pi r}\left(\frac{\pi}{2}-1\right)$

(3) $\frac{\mu_0 I}{4\pi r}\left(\frac{3\pi}{2}+1\right)$

(4) $\frac{\mu_0 I}{4\pi r}\left(\frac{3\pi}{2}+2\right)$

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12. A long straight wire in the horizontal plane carries a current of 50 A in north to south direction. Give the magnitude and direction of B at a point 2.5m east of the wire :-

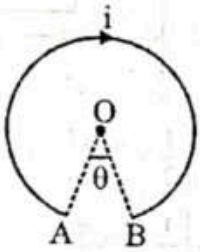
(1) 4×10^{-6} T upwards

(2) 6×10^{-6} T down wards

(3) 4×10^{-6} T east wards

(4) Both (1) and (3)

13. A current carrying wire AB of the length L is turned along a circle, as shown in figure. The magnetic field at the centre O.



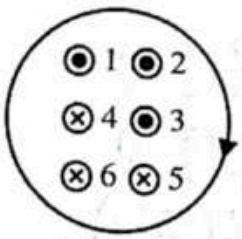
(1) $\frac{\mu_0 \pi i}{L}\left(\frac{2\pi-\theta}{2\pi}\right)^2$

(2) $\frac{\mu_0 \pi i}{L}\left(\frac{2\pi-\theta}{2\pi}\right)$

(3) $\frac{\mu_0 \pi i}{L}(2\pi-\theta)$

(4) $\frac{\mu_0 \pi i}{L}(2\pi+\theta)^2$

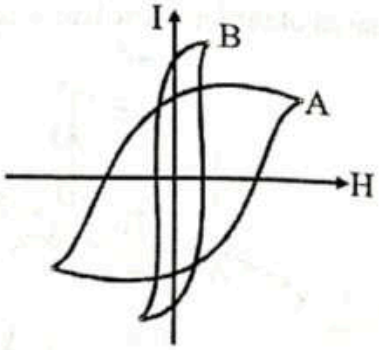
14. Six long wires of current $I_1 = 1$ A, $I_2 = 2$A, $I_3 = 3$A, $I_4 = 1$A, $I_5 = 5$A and $I_6 = 4$A cut the page perpendicular at the points 1,2,3,4,5 and 6 respectively as shown in the figure. Find the value of the integral $\int \vec{B}.d\vec{l}$ around the closed path.



(1) $2\mu_0$ weber m^{-1} (2) $4\mu_0$ weber m^{-1}

(3) $6\mu_0$ weber m^{-1} (4) $8\mu_0$ weber m^{-1}

15. The intensity of magnetisation I is plotted against the magnetising field H for different substances. The curves A and B shown in the adjoining figure are associated with :-



(1) Soft iron and steel respectively

(2) Steel and soft iron respectively

(3) A diamagnetic and paramagnetic substance respectively

(4) A paramagnetic and diamagnetic substance respectively

16. Match **List-I** with **List-II**

	List I (Material)		**List II (Example)**
A.	Diamagnet	I.	Alnico
B.	Paramagnet	II.	Copper
C.	Soft ferromagnet	III.	Aluminium
D.	Hard ferromagnet	IV.	Gadolinium

Choose the **correct** answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-III, B-II, C-I, D-IV

(3) A-II, B-III, C-IV, D-I

(4) A-I, B-II, C-III, D-IV

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17. A solenoid has core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2A. If the number of turns is 1000 per metre, calculate H (magnetising field) and I (Intensity of magnetization).

(1) $2 \times 10^3\ Am^{-1}$, $798 \times 10^3\ Am^{-1}$

(2) $1.0\ Am^{-1}$, $1.5 \times 10^5\ Am^{-1}$

(3) $6.8 \times 10^{-5}\ Am^{-1}$, $1.2 \times 10^{-5}\ Am^{-1}$

(4) $2.1 \times 10^{-4}\ Am^{-1}$, $-2.6 \times 10^5\ Am^{-1}$

18. If the angular momentum of an electron is $\vec{J}$ then the magnitude of the magnetic moment will be:-

(1) $\frac{eJ}{m}$

(2) $\frac{eJ}{2m}$

(3) eJ 2m

(4) $\frac{2m}{eJ}$

19. For a series RLC circuit $R = X_L = 2X_C$. The impedance of the circuit and phase difference between V and i will be :-

(1) $\frac{\sqrt{5}R}{2}, \tan^{-1}(2)$

(2) $\frac{\sqrt{5}R}{2}, \tan^{-1}\left(\frac{1}{2}\right)$

(3) $\sqrt{5}X_C, \tan^{-1}(2)$

(4) $\sqrt{5}R, \tan^{-1}\left(\frac{1}{2}\right)$

20. The voltage supplied to a circuit is given by $V = V_0 t^{3/2}$, where t is time in second. Find the RMS value of voltage for the period t = 0 to t = 1s :-

(1) $\frac{V_0}{2}$ (2) V_0

(3) $\frac{3V_0}{2}$ (4) $\frac{V_0}{4}$

21. When a voltage $v_s = 200\sqrt{2}\sin(\omega t + 15°)$ is applied to an AC circuit then current in the circuit is found to be $i = 2\sin(\omega t + \pi/4)$. Average power consumed in the circuit will be :-

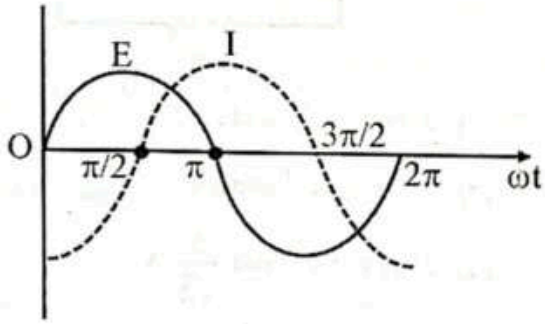
(1) 200 watt

(2) $400\sqrt{2}$ watt

(3) $100\sqrt{6}$ watt

(4) $200\sqrt{2}$ watt

22. The variation of the instantaneous current (I) and the instantaneous emf (E) with phase (ωt) in a circuit is as shown in fig. Which of the following statement is correct –



(1) The voltage lags behind the current by $\pi/2$

(2) The voltage leads the current by $\pi/2$

(3) The voltage and the current are in phase

(4) The voltage leads the current by π

23. Which of the following combination should be selected for better tuning of an L-C-R circuit used for communication?

(1) $R = 20\Omega$, $L = 1.5$ H, $C = 35\mu F$

(2) $R = 25\Omega$, $L = 2.5$ H, $C = 45\mu F$

(3) $R = 15\Omega$, $L = 3.5$ H, $C = 30\mu F$

(4) $R = 25\Omega$, $L = 1.5$ H, $C = 45\mu F$

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24. A bulb and a capacitor are connected in series to a source of alternating current. If its frequency is increased, while keeping the voltage of the source constant, then

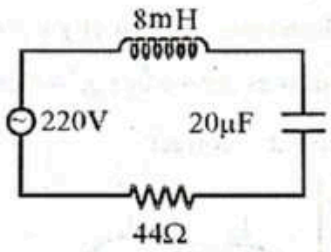
(1) Bulb will give more intense light

(2) Bulb will give less intense light

(3) Bulb will give light of same intensity as before

(4) Bulb will stop radiating light

25. For the series LCR circuit shown in the figure, what is the resonance frequency and the amplitude of the current at the resonating frequency :-



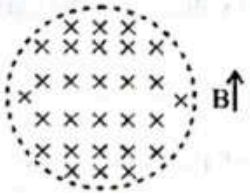
(1) 2500 rad s^{-1} and $5\sqrt{2}$ A

(2) 2500 rad s^{-1} and 5A

(3) 2500 rad s^{-1} and $\frac{5}{\sqrt{2}}$ A

(4) 25 rad s^{-1} and $5\sqrt{2}$ A

26. In this given figure if magnetic field increases with time then pattern of induced electric field lines will be :-



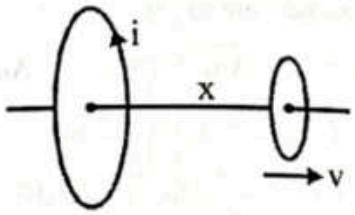
(1) A.C.W. concentric circular field lines in the plane of the paper

(2) C.W. concentric circular field lines in the plane of the paper

(3) A.C.W. concentric circular field lines, perpendicular to the plane of the paper

(4) C.W. concentric circular field lines, perpendicular to the plane of the paper

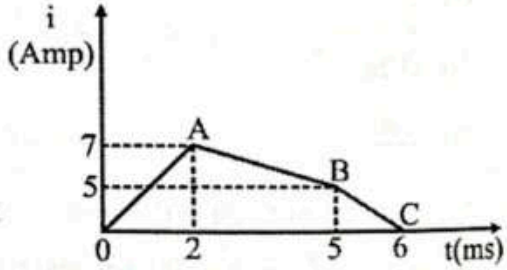
27. Diagram shows two parallel and coaxial loops, smaller loop is of radius r. Magnetic field due to current 'i' in larger loop (radius R >> r) is nearly constant throughout the smaller loop. Separation between the loops is x (>> R), then mutual inductance between the loops :-



(1) $\frac{3\mu_0\pi R^2r^2}{2x^3}$ (2) $\frac{4\mu_0\pi R^2r^2}{6x^3}$

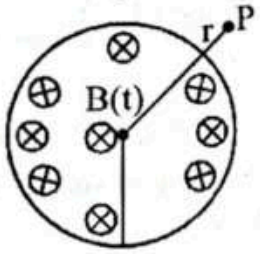
(3) $\frac{\mu_0\pi R^2r^2}{4x^3}$ (4) $\frac{\mu_0\pi R^2r^2}{2x^3}$

28. The current through a 6 mH inductor is shown in the following graph. The induced emf at t = 4 ms will be :-



(1) 3V (2) −3V (3) 4V (4) −4V

29. A uniform but time-varying magnetic field B(t) exists within a circular region of radius 'a' and is directed into the plane of the paper as shown in figure. The magnitude of the induced electric field at the point P at a distance 'r' from the centre of the circular region :-

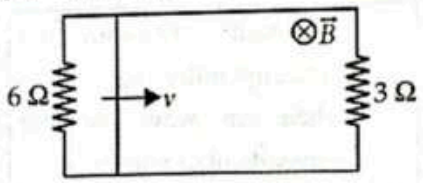


(1) Is zero (2) $E \propto \frac{1}{r}$

(3) $E \propto r$ (4) $E \propto \frac{1}{r^2}$

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30. A rectangular loop with a sliding connector of length $\ell = 1.0m$ is situated in a uniform magnetic field B = 2T, perpendicular to the plane of the loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in figure. The external force required to keep the connector moving with a constant velocity $v = 2ms^{-1}$ is



(1) 2N (2) 1N

(3) 4N (4) 6N

31. **Statement-I :** When two coils are wound on each other, the mutual induction between the coils is maximum.

Statement-II : Mutual induction does not depend on the relative orientation of the coils.

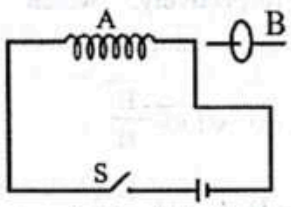
(1) Both Statement-I & Statement-II are True & the Statement-II is a correct explanation of the Statement-I.

(2) Both Statement-I & Statement-II are True but Statement-II is not correct explanation of the Statement-I.

(3) Statement-I is True but the Statement-II is False.

(4) Both Statement-I & Statement-II are False.

32.



As the switch S is closed at t = 0 the ring B :-

(1) is attracted towards A

(2) is repelled by A

(3) is initially repelled and then attracted

(4) is initially attracted and then repelled

33. Consider two identical, coaxial circular loops carrying a current i each, circulating in the same direction. If the loops approach each other :

(1) The current in each loop will decrease

(2) The current in each loop will increase

(3) The current in each loop will remain the same

(4) The current in one loop will increase and in the other loop will decrease

34. The current through an inductor of 1H is given by i = 3t sin t. The Induced voltage across the inductor:-

(1) 3sint + 3cost (2) − 3cost − t sint

(3) − 3sint − 3t cost (4) 3t cost + sint

35. The resistance and inductance of series circuit are 5 Ω and 20 H respectively. At the instant of closing the key, the current is increasing at the rate 4 A/sec. The supply voltage

(1) 20 V (2) 80 V

(3) 100 V (4) 120 V

36. A vertical bar magnet is dropped from position on the axis of a fixed metallic coil as shown in fig - I. In fig - II the magnet is fixed and horizontal coil is dropped. The acceleration of the magnet and coil are a_1 and a_2 respectively then

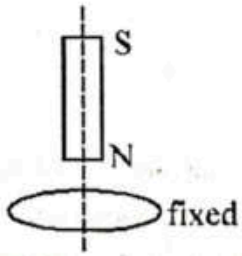


fig. I

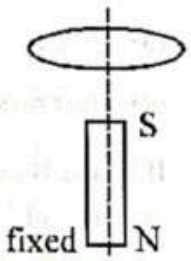


fig.II

(1) $a_1 > g, a_2 > g$ (2) $a_1 > g, a_2 < g$

(3) $a_1 < g, a_2 < g$ (4) $a_1 < g, a_2 > g$

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37. The conduction current is same as displacement current when source is

(1) ac only (2) dc only

(3) either ac or dc (4) neither dc nor ac

38. Which one among the following does not represent Maxwell's equation?

(1) $\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$

(2) $\oint \vec{B}.d\vec{A} = 0$

(3) $\oint \vec{E}.d\vec{\ell} = \frac{-dB}{dt}$

(4) $\oint \vec{B}.d\vec{\ell} = \mu_0 I_C + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$

39. Electromagnetic wave consists of periodically oscillating electric and magnetic vectors

(1) in mutually perpendicular planes but vibrating with a phase difference of π

(2) in mutually perpendicular planes but vibrating with a phase difference of $\frac{\pi}{2}$

(3) in randomly oriented planes but vibrating in phase

(4) in mutually perpendicular planes but vibrating in same phase.

40. Which of the following rays is not an electromagnetic wave ?

(1) X-rays

(2) γ-rays

(3) β-rays

(4) Heat rays

41. If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of the electromagnetic wave, the direction of propagation of electromagnetic wave is along

(1) $\vec{E}$ (2) $\vec{B}$

(3) $\vec{B} \times \vec{E}$ (4) $\vec{E} \times \vec{B}$

42. Match Statement of column I with expression of column II

Column I		Column II	
(A)	In em wave, the rate of flow of energy per unit area	(P)	$\frac{B_0^2}{2\mu_0}$
(B)	Average energy density of the electromagnetic wave	(Q)	$\frac{E_0 B_0}{\mu_0 c}$
(C)	Radiation pressure on perfectly reflecting surface when em wave incident perpendicularly on it	(R)	$\frac{E_0 B_0}{2\mu_0 c}$
(D)	Radiation pressure on absorbing surface when em wave is incident on it	(S)	$\frac{cB^2}{\mu_0}$

(1) (A)→(P); (B)→(Q); (C)→(R); (D)→(S)

(2) (A)→(S); (B)→(P); (C)→(Q); (D)→(R)

(3) (A)→(S); (B)→(P); (C)→(R); (D)→(Q)

(4) (A)→(Q); (B)→(R); (C)→(S); (D)→(P)

43. The electric field associated with an electromagnetic wave in vacuum is given by $\vec{E} = 40 \cos (kz - 6 \times 10^8 t)\hat{i}$, where E, z and t are in volt per metre, metre and second respectively. The value of wave vector k is

(1) $2\ m^{-1}$ (2) $0.5\ m^{-1}$

(3) $6\ m^{-1}$ (4) $3\ m^{-1}$

44. If E and B denote electric and magnetic fields respectively, which of the following is dimensionless?

(1) $\sqrt{\mu_0 \varepsilon_0}\frac{E}{B}$ (2) $\mu_0 \varepsilon_0 \frac{E}{B}$

(3) $\mu_0 \varepsilon_0 \left(\frac{B}{E}\right)^2$ (4) $\frac{E}{\varepsilon_0}\frac{\mu_0}{B}$

45. For L-C-R a.c. circuit resonance frequency is 600 Hz and frequencies at half power points are 550 Hz and 650 Hz. What will be the Q - factor?

(1) $\frac{1}{6}$ (2) $\frac{1}{3}$ (3) 6 (4) 3

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SUBJECT : CHEMISTRY

Topic : SYLLABUS-5.

46. In which of the following pair, the EAN of central metal atom is not same :-

(1) $[Fe(CN)_6]^{-3}$ and $[Fe(NH_3)_6]^{+3}$

(2) $[Cr(NH_3)_6]^{+3}$ and $[Cr(CN)_6]^{-3}$

(3) $[FeF_6]^{-3}$ and $[Fe(CN)_6]^{-3}$

(4) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{-2}$

47. **Assertion :** BF_3 is planar due to back bonding.
Reason : BF_3 has no reactivity towards water at room temperature.

(1) Both **Assertion** & **Reason** are true & the **Reason** is a correct explanation of the **Assertion**.

(2) Both **Assertion** & **Reason** are true but **Reason** is not a correct explanation of the **Assertion**.

(3) **Assertion** is true but the **Reason** is False.

(4) Both **Assertion** & **Reason** are False.

48. Complex

(a) $Na_2[Fe(CN)_5NO]$ (b)$[Fe(H_2O)_5NO]SO_4$

(c) $K_4[Fe(CN)_6]$

Which of the following statement are wrong ?

(1) (a) is used in detection of S^{2-}

(2) complex part of each compound is octahedral

(3) (c) is used in detection of Fe^{3+}

(4) Oxidation state of Fe in (a) and (b) is +1

49. The total No. of possible isomers of the compound $[Cu^{II}(NH_3)_4][Pt^{II}Cl_4]$ are

(1) 3 (2) 5 (3) 4 (4) 6

50. **Assertion :** SiO_2 is not isostructural with CO_2.
Reason : $2p\pi$-$3p\pi$ colateral overlapping in SiO_2 is not as effective as $2p\pi$-$2p\pi$ in CO_2.

(1) Both **Assertion** & **Reason** are true & the **Reason** is a correct explanation of the **Assertion**.

(2) Both **Assertion** & **Reason** are true but **Reason** is not a correct explanation of the **Assertion**.

(3) **Assertion** is true but the **Reason** is False.

(4) Both **Assertion** & **Reason** are False.

51. Select the correct statement from the following.

(1) $[Sc(H_2O)_6]^{3+}$ and $[Ti(H_2O)_6]^{3+}$ both are colourless.

(2) $[Co(NH_3)_4Br_2]Cl$ shows ionization isomers and geometrical isomers.

(3) $[Pd(NO_2)_2(NH_3)_2]$ is square planar and shows geometrical as well as linkage isomers.

(4) Both (2) and (3) are correct.

52. **Assertion :** The trans isomer of complex $[CoCl_2(en)_2]^{\oplus}$ is optically active.
Reason : The cis isomer of complex $[CoCl_2(en)_2]^{\oplus}$ has a plane of symmetry.

(1) Both **Assertion** & **Reason** are true & the **Reason** is a correct explanation of the **Assertion**.

(2) Both **Assertion** & **Reason** are true but **Reason** is not a correct explanation of the **Assertion**.

(3) **Assertion** is true but the **Reason** is False.

(4) Both **Assertion** & **Reason** are False.

53. What is the common formula of Turnbull's blue and Prussian blue ?

(1) $Fe_3[Fe(CN)_6]_2$ (2) $Fe_4[Fe(CN)_6]_3$

(3) $KFe[Fe(CN)_6]$ (4) $KFe_2[Fe(CN)_6]$

54. $XeF_6 + SbF_5 \longrightarrow \underset{\text{Cation}}{A} + \underset{\text{Anion}}{B}$

Incorrect statement for above change is :-

(1) XeF_6 act as F^- donor

(2) Hybridisation state of central atom in cationic part is sp^3d^2

(3) Anionic part is isostructural with PCl_6^-

(4) Both cation & anion have identical shape

55. Choose correct statement :-

(1) $[Co(CO)_5NH_3]^{+2}$ is a $(\sigma + \pi)$ bonded organo metallic compound.

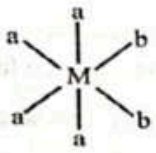
(2) $[Pt(NH_3)_4Cl_2]Br_2$ and $[Pt(NH_3)_4Br_2]Cl_2$ are coordination isomers to each other.

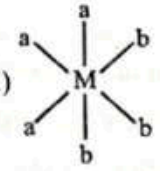
(3) High spin d^4 octahedral complex CFSE is $-0.6\,\Delta_0 + P$

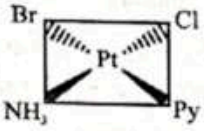
(4) Coordination No. of Fe in $[Fe(\eta^5-C_5H_5)_2]$ is 4

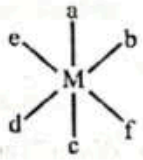
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56. Select optically active complex.

(I) 

(II) 

(III) 

(IV) 

(1) Only (IV) (2) (I) and (IV)

(3) (I), (II) and (III) (4) (I) and (III)

57. Match column I & II & select correct relation :-

Column-I		Column-II	
(A)	Estimation of hardness of water	(i)	Cis-Platin
(B)	Detection of nickel	(ii)	EDTA
(C)	Electro plating	(iii)	DMG
(D)	Treatment of cancer	(iv)	Potasium Dicyanoargante(I)

(1) (A)-(i), (B)-(iv), (C)-(ii), (D)-(iii)

(2) (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)

(3) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)

(4) (A)-(iv), (B)-(ii), (C)-(iii), (D)-(i)

58. Choose the correct matching :-

	Column-I (Given Mixture)		Column-II (Colour Precipitate)
(P)	Pb^{+2} + dil HCl + $H_2S \rightarrow$	(A)	Rosy Red ppt.
(Q)	Fe^{+3} + NaOH $\rightarrow$	(B)	White ppt.
(R)	Ni^{+2} + DMG solution $\rightarrow$	(C)	Black ppt.
(S)	Zn^{+2} + KCN solution $\rightarrow$	(D)	Reddish-Brown ppt.

(1) P-C, Q-D, R-A, S-B (2) P-C, Q-D, R-B, S-A

(3) P-B, Q-C, R-A, S-D (4) P-D, Q-C, R-A, S-B

59. Which compounds does not exist ?

(1) MnF_6 (2) MnF_4

(3) MnO_3F (4) MnO_4^{-2}

60. In the precipitation of the iron group in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to :

(1) decrease concentration of OH^- ions

(2) prevent interference by phosphate ions

(3) increase concentration of Cl^- ions

(4) increase concentration of NH_4^+ ions

61. **Statement-I :** The chlorides of Be^{2+} and Mg^{2+} do not impart any colour to the flame test.

Statement-II : The hydration energy of Be^{2+} is higher than that of Mg^{2+}.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

62. In which reaction no color change will be observed ?

(1) $K_2Cr_2O_7 \xrightarrow[H^+]{CO_2}$

(2) $K_2Cr_2O_7 \xrightarrow[H^+]{SO_2}$

(3) $Na_2S \xrightarrow{NA_2[Fe(CN)_5NO]}$

(4) All of these

63. Match list-I with List-II

	List-I (Precipiteting reagent and conditions)		List-II (Cation)
A	$NH_4Cl + NH_4OH$	I	Mn^{2+}
B	$NH_4OH + (NH_4)_2CO_3$	II	Pb^{2+}
C	$NH_4OH + NH_4Cl + H_2S$ gas	III	Al^{3+}
D	dilute HCl	IV	Sr^{2+}

(1) A - III, B - IV, C - I, D - II

(2) A - III, B - IV, C - II, D - I

(3) A - IV, B - III, C - II, D - I

(4) A - IV, B - III, C - I, D - II

64. H_2S on passing through ammonical solution gives white ppt. The white ppt. is of :

(1) CoS (2) NiS (3) MnS (4) ZnS

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65. Which of following is correct matching :-

(1) I_2Cl_6 ⟶ Planar molecule

(2) $BeCl_2$ ⟶ Polymer in solid phase

(3) In B_2H_6, All Hydrogen atom's are not present in a same plane.

(4) All are correct

66. How many P–O–P bonds appear in cyclic trimer of metaphosphoric acid :-

(1) 4 (2) 3 (3) 2 (4) 5

67. The compound which does not contain peroxy linkage :-

(1) H_2SO_5 (2) H_3PO_5 (3) $H_2S_2O_6$ (4) $H_2S_2O_8$

68. Total number of sulphur atom in $H_2S_nO_6$ which form covalent bond in their zero oxidation state ?

(1) n (2) n – 1 (3) n – 2 (4) n – 3

69. Select correct information regarding $B_3N_3H_6$:

(1) $B_3N_3H_6$ is non-planar compound

(2) $B_3N_3H_6$ is Non-polar compound

(3) In $B_3N_3H_6$ back bonding in not available

(4) $B_3N_3H_6$ is less reactive than C_6H_6

70. Which alkali metal does not forms any type of Alum?

(1) K (2) Na (3) Cs (4) Li

71. Pick the incorrect option from the choices given below.

(1) $H_3PO_2 > H_3PO_3$: Reducing nature

(2) $H_3PO_4 > H_3PO_2$: Basicity

(3) H_3PO_3 : Disproportionates on heating

(4) $H_4P_2O_7$ can act as reducing agent

72. In which group electronegativity regularly decrease?

(1) Group–14 (2) Group–13

(3) Group–17 (4) All of these

73. The incorrect reaction among the given below reaction is –

(1) Fe + Conc. HCl ⟶ $FeCl_3 + H_2$

(2) $XeF_4 + H_2O \longrightarrow Xe + XeO_3 + O_2 + HF$

(3) $SiCl_4 + H_2O \longrightarrow Si(OH)_4 + HCl$

(4) NaF + Conc. $H_2SO_4 \longrightarrow HF + NaHSO_4$

74. In which of the following compound, central atom not used vacant orbital -

(1) B_2H_6 (2) $BeCl_2$ (Solid)

(3) C_2H_6 (4) Al_2Cl_6

75. During Borax Bead test Cr^{+3} gives color :-

(1) Green (2) Violet (3) Brown (4) Pink

76. Incorrect about allotrope of sulphur :-

(1) Total No. of lone pair in S_8 are eight

(2) Both rhombic and monoclinic have S_8 molecules

(3) S_8 ring is crown shaped

(4) S_2 is paramagnetic like O_2

77. Which of the following is incorrect for H_3BO_3 :-

(1) It has a layer structure in which H_3BO_3 unit are joined by H-bonds

(2) It does not act as a proton donor but acts as an acid by accepting hydroxyl ion

(3) It can show back bonding.

(4) Basicity of H_3BO_3 is three.

78. Which of the following statement is correct ? [Statement T for True and F for False]

(i) Most stable trihalide of 'N' family is NF_3.

(ii) Most stable pentahalide of 'Bi' is BiF_5

(iii) Most stable allotropes of 'N' is black form

(iv) Most stable Hydride of 'N' family is NH_3

(1) T T F T (2) T T F F

(3) T F F T (4) F F F F

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79. **Assertion (A) :** Drunken driver test done by using of acidic $K_2Cr_2O_7$.
Reason (R) : $Na_2Cr_2O_7$ is a non-Hygroscopic and not use in volumetric analysis.
(1) Both **(A)** & **(R)** are true and **(R)** is a correct explanation of the **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false
(4) **(A)** is false but **(R)** is true

80. Which of the following statement is correct ?
(1) In nitrogen family all the elements show allotropy
(2) In the case of nitrogen all oxidation states from +1 to +5 tend to disproportionate in both alkali and acidic solution
(3) Due to small size and high electronegativity, nitrogen has stronger catenation tendency than phosphorous
(4) Trihalides of pnicogen family except those of nitrogen are stable

81. Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest oxidising character ($E^0_{M^{3+}/M^{2+}}$) ?
(1) Cr(Z = 24) (2) Mn(Z = 25)
(3) Fe (Z = 26) (4) Co(Z = 27)

82. When hydrogen peroxide is added to acidified potassium dichromate, a blue colour is produced due to formation of :-
(1) CrO_3 (2) $Cr_2O_7^{-2}$
(3) CrO_5 (4) CrO_4^{-2}

83. $KMnO_4$ is strong oxidizing agent in acidic medium. To provide acidic medium H_2SO_4 is used instead of HCl. This is because :-
(1) H_2SO_4 is a stronger acid than HCl
(2) HCl is oxidized by $KMnO_4$ into Cl_2
(3) H_2SO_4 is a dibasic acid
(4) Rate is faster in the presence of H_2SO_4

84. Incorrect order of oxidising strength :-
(1) $GeO_2 < SnO_2 < PbO_2$
(2) $HOCl < HClO_2 < HClO_4$
(3) $MnO_4^- > TcO_4^- > ReO_4^-$
(4) $MnO_4^- > Cr_2O_7^{-2} > VO_2^+$

85. Select the incorrect statement about $K_2Cr_2O_7$:-
(1) the bond angle of $Cr-\widehat{O}-Cr > 109.5$
(2) Maximum No. of identical Cr–O bond's are six in $Cr_2O_7^{-2}$
(3) two types of Cr–O bond length are present
(4) It act as reducing agent in acidic medium

86. Select the correct statement ?
(1) Yb, Eu can show +2 oxidation state
(2) Ce, Tb, can show +4 oxidation state
(3) La, Lu, Gd show abnormally low value of 3rd Ionisation enthalpy.
(4) All are correct statement

87. In fusion of chromite Ore with sodium Carbonate in Excess of Air which of the following is not formed?
(1) Fe_2O_3 (2) Na_2CrO_4
(3) CO_2 (4) $Na_2Cr_2O_7$

88. Which statement is wrong :
(1) Reactivity of all lanthanoid's similar to aluminium
(2) Ce^{+4} is a good analytical reagent because of reaction rate is very slow
(3) Mg based alloy is used in lighter flints
(4) Yb^{2+} is a strong reducing agent

89. MnO_2 dissolves in concentrated HCl to form :
(1) Mn^{4+} ion and Cl_2 (2) Mn^{2+} ion and Cl_2
(3) $[MnCl_4]^{2-}$ and Cl_2 (4) Only $[MnCl_4]^{2-}$

90. False statement about $KMnO_4$ is-
(1) it is coloured due to charge transfer spectra
(2) it shows temperature dependent weak paramagnetism
(3) $p\pi$-$p\pi$ and $p\pi$-$d\pi$ both type of bonds are present
(4) All Mn - O bonds are identical

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SUBJECT : BIOLOGY

Topic : SYLLABUS-5.

91. Identify the traits (A-D) chosen by Mendel.

Characters in Pod (Chosen by Mendel)	Dominant	Recessive
1. Colour	A	B
2. Shape	C	D

	A	B	C	D
1.	Green	Yellow	Inflated	Constricted
2.	Yellow	Green	Inflated	Constricted
3.	Green	Yellow	Constricted	Inflated
4.	Yellow	Green	Constricted	Inflated

92. Which of the following characters are not suitable for genetical studies of drosophila ?

(a) They could be grown on simple synthetic medium in laboratory.

(b) They complete their life cycle in about 2-weeks.

(c) Single mating produces few number of progeny flies.

(d) They have many types of hereditary variations that can be seen with low power microscope.

(e) Male & Female flies are not easily distinguishable.

(1) a, b, c

(2) a, b, c, d, e

(3) d and e

(4) c and e

93. Recombination frequency between the genes a & c is 3%, b & c is 7%, b & d is 2%, a & b is 10% and a & d is 12%. What will be the distance between genes c & d ?

(1) 5 unit (2) 8 unit

(3) 9 unit (4) 20 unit

94. Consider the following statements and choose the correct options:

(A) In incomplete dominance, the F_1 had a phenotype that did not resemble either of the two parents and was in between the two.

(B) In co-dominance, there is blending of characters.

(C) Dominance is an autonomous feature of a gene or the product that it has information for.

(D) In an individual only two alleles can be present, multiple alleles can be found only when population studies are made.

(E) Gene which controls more than one character is called multiple alleles.

(1) A, B and E

(2) B, C and D

(3) A and D

(4) B and D

95. In an example of incomplete dominance, pure red flowered plants are crossed with pure white flowered plants and the F_1 individuals have pink flowers. Which one of the following is **not correct** ?

(1) Pink flowered plant will produce only pink flowered offsprings.

(2) Pink flowered plant will produce three types of offsprings if self pollinated.

(3) The genes of the hybrid pink flowered plant will segregate, if self pollinated.

(4) Half of the offsprings of the pink flowered plant will be homozygous, if self-pollinated.

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96. Find out the correct match from column I, II and III of following table :-

Column I Cross type		Column II No. of genotypes		Column III No. of phenotypes	
(A)	Monohybrid	(I)	9	(1)	8
(B)	Dihybrid	(II)	27	(2)	4
(C)	Trihybrid	(III)	3	(3)	2

(1) A – III – 3, B – I – 2, C – II – 1
(2) A – II – 1, B – III – 2, C – I – 3
(3) A – I – 2, B – III – 1, C – II – 3
(4) A – III – 3, B – II – 1, C – I – 2

97. **Assertion** : Study of family history about inheritance of particular trait provide an alternative of control crosses.
Reason : Control crosses that can be performed in pea plants are not possible in case of human beings.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

98. **Statement-I** : Mendel self pollinated dwarf F_2 plants and found that they continued to generate tall plants in F_3 and F_4 generations.
Statement-II : In a typical test cross an organism showing a dominant phenotype (whose genotype is to be determined) is crossed with recessive parent.

(1) Both statement-I & Statement-II are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

99. If the genotypes of husband and wife are I^AI^B and I^AI^O : Among the blood types of their children, how many different genotypes and phenotypes are possible?

(1) 2 genotypes, 3 phenotypes
(2) 3 genotypes, 4 phenotypes
(3) 4 genotypes, 4 phenotypes
(4) 4 genotypes, 3 phenotypes

100. Read the following statements and choose the correct option :-
(A) Simply looking at the phenotype of a dominant trait, it is impossible to know the genotypic composition.
(B) Mendel selected two characters of seed for study.
(C) On the basis of monohybrid crosses, Mendel proposed only one conclusion.

(1) A & B are correct & C is incorrect
(2) A is incorrect
(3) A, B & C are correct
(4) Only A is correct

101. The correct match is :-

Column-A		Column-B	
I.	ABO blood groups	A	Dihybrid cross
II.	Law of segregation	B	Monohybrid cross
III.	Law of independent assortment	C	Quantitative inheritance
IV.	Polygenic inheritance	D	Multiple allelism

(1) I-B, II-C, III-D, IV-A
(2) I-D, II-A, III-B, IV-C
(3) I-D, II-B, III-A, IV-C
(4) I-B, II-A, III-D, IV-C

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102. **Assertion (A) :** Sutton and Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes.

Reason (R) : Frequency of recombination between gene pairs on same chromosomes is used as a measure of distance between genes.

(1) A and R both correct but R is not correct explanation of A.

(2) A and R both correct and R is correct explanation of A.

(3) A is correct but R is incorrect.

(4) A is incorrect but R is correct.

103. Which of the following conclusions of Mendel can not be explained by monohybrid cross ?

(1) Dominance

(2) Segregation

(3) Independent assortment

(4) All the above

104. Which of the following statement is incorrect?

(1) Test cross helps to find out phenotype of unknown dominant.

(2) Linkage is the exception of law of independent assortment.

(3) According to law of dominance factors occurs in pair.

(4) Hugo de vries did experiment on evening primrose.

105. Which of the following is a chromosomal disorder ?

(1) Colour blindness (2) Down's syndrome

(3) Haemophilia (4) Alkaptonuria

106. Choose the correct match :

(1) Monohybrid cross phenotypic ratio $\longrightarrow$ 9 : 3 : 3 : 1

(2) Dihybrid cross phenotypic ratio $\longrightarrow$ 1 : 2 : 1

(3) Monohybrid cross genotypic ratio $\longrightarrow$ 3 : 1

(4) Dihybrid cross genotypic ratio $\longrightarrow$ 1:2:1:2:4:2:1:2:1

107. A heterozygous round seed shape garden pea plant was crossed with wrinkled seed shape plant. Round is dominant over wrinkled. Which one of the following represents the expected combinations among 60 progenies formed ?

(1) 30 produced round and 30 produced wrinkled seed

(2) 45 produced round and 15 produced wrinkled seed

(3) All 60 produced round seed

(4) All 60 produced wrinkled seed

108. Morgan and his group found that in *Drosophila* the genes eye colour and body colour were :-

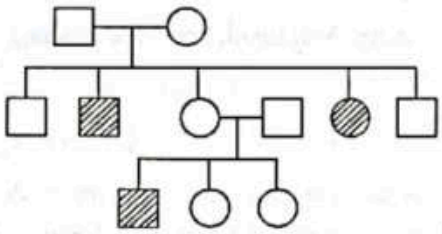
(1) Present on autosomes

(2) Tightly linked genes

(3) Loosely linked genes

(4) Both (1) and (2)

109. Given pedigree chart shown inheritance of :-



(1) Myotonic dystrophy

(2) Sickle cell Anaemia

(3) Colour blindness

(4) Haemophilia

110. In which of the following F_1 has a phenotype that did not resemble either of the two parents and was in between the two ?

(1) Flower colour in garden pea

(2) Size of starch grain in pea

(3) Flower colour in snapdragon

(4) Both (2) and (3)

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111. Identify A and B in following statements -
Non-parental combinations in F_2 generation of dihybrid crosses obtained by Mendel is due to ___A___, and by Morgan is due to ___B___ .

(1) A - Segregation ; B - Crossing over

(2) A - Non-disjunction ; B - Linkage

(3) A - Independent Assortment ; B - Crossing-over

(4) A - Crossing over ; Mutation

112. Which of the following statement are correct ?
(a) Replication of DNA on template strand with polarity 5'⟶3' is continuous.
(b) Replication of DNA on template strand with polarity 3'⟶5' is discontinuous.
(c) Deoxyribonucleoside triphosphates act as substrate as well as provide energy during replication of DNA.
(d) Meselson and stahl got two heavy and two light DNA in 2nd generation of *E.coli.*
(e) The unequivocal proof that DNA is the genetic material came from the experiments of Avery, MacLeod, and McCarty in 1952.
Option :

(1) a, b, c (2) d, e (3) a, b, e (4) Only c

113. Select the correct statements with reference to experiments of Avery, MacLeod, and McCarty.
(A) DNA from S-bacteria caused R-bacteria to become transformed.
(B) Protein-digesting enzymes (proteases) did affect transformation.
(C) RNA-digesting enzymes (RNases) did not affect transformation.
(D) DNA-digesting enzymes (DNases) did not inhibit transformation.
(E) DNA from R-bacteria caused S-bacteria to become transformed.
Choose the most appropriate answer from the options given below :-

(1) B and E only (2) A and B only

(3) A and C only (4) A and D only

114. Read the following terminology of Column-I and Column-II and identify their correct match.

Column-I		Column-II	
A.	Building blocks of DNA	(i)	Helical
B.	Genetic code is composed of	(ii)	Transcription
C.	Shape of DNA molecule	(iii)	Translation
D.	Process of formation of RNA from DNA is called	(iv)	Nucleotide
		(v)	3 base
		(vi)	Transformation
		(vii)	Circular

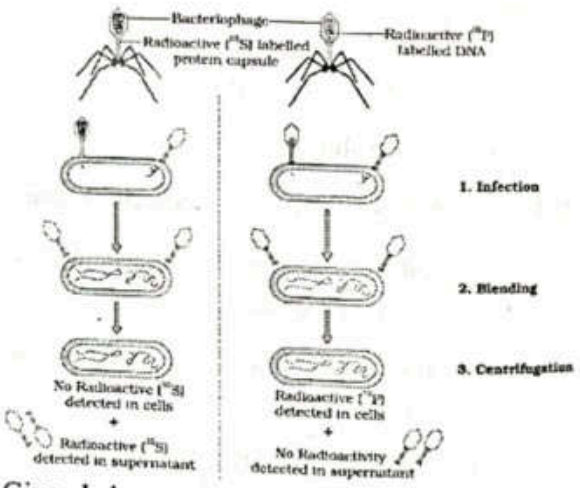
(1) A-iv, B-v, C-vii, D-iii

(2) A-iv, B-v, C-i, D-ii

(3) A-i, B-v, C-vii, D-ii

(4) A-i, B-vi, C-vii, D-iii

115.



Given below are two statements about the above experiment.

Statement-I : The Experiment was performed by Avery, MacLeod and McCarty.

Statement-II : This experiment indicates that protein did not enter the bacteria from the viruses.

(1) Statement-I is correct but statement-II is incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Both statement-I and statement-II are incorrect.

(4) Both statement-I and statement-II are correct.

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116. **Statement-I** : Transcription in prokaryotes always proceeds undirectionally from the promoter region to terminator region, synthesizing RNA in 3'-5' direction using ribonucleoside triphosphate as substrate.

Statement-II : The sigma factor dissociates after initiation and is not required for elongation or termination.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I and Statement-II are correct.

(4) Statement-I is incorrect, Statement-II is correct

117. Given below are few examples of chemical bonds and interaction :-

Peptide bond, disulphide bond, glycosidic bond, hydrogen bonds, phosphodiester bonds, phosphoester bonds.

How many of the above examples are present in a double stranded DNA ?

(1) Only three (2) Only four

(3) Only two (4) Six

118. Choose correct statements regarding double helix structure of DNA ?

(a) Antiparallel polarity

(b) Double strand

(c) Uniform diameter

(d) The pitch of the helix is 3.4 Å

(1) a, b only (2) a, b and c only

(3) a and d only (4) a, b, c and d

119. From the following identify the correct combination of salient features of the genetic code.

(1) Universal, Non-ambiguous, Overlapping

(2) Degenerate, Over lapping, commaless

(3) Universal, Unambiguous, Degenerate

(4) Degenerate, Overlapping, Non-ambiguous

120. **Statement-I** : In the experiment of Meselson and stall amount of hybrid DNA after 20 minute and 40 minute was 100% and 50% respectively.

Statement-II : Energetically DNA replication is very expensive process.

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-II is correct but statement-I is incorrect.

(4) Both statement-I and statement-II are incorrect.

121. **Assertion(A)** : DNA chemically is less reactive and structurally more stable as compared to RNA.

Reason(R) : DNA dependent DNA polymerase uses a DNA template to catalyse the polymerisation of deoxynucleotides.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(3) (A) is correct but (R) is incorrect.

(4) Both (A) and (R) are incorrect.

122. Match the following columns :

	Column-I		Column-II
(A)	5'-AUG-3'	I.	Segment of DNA
(B)	RNA with introns and exon	II.	Chromatin
(C)	Gene	III.	hnRNA
(D)	Nucleosomes	IV.	Initiation codon

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-II, C-I, D-III

(3) A-II, B-I, C-IV, D-III

(4) A-II, B-III, C-I, D-IV

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123. How many statements are incorrect ?

(i) The backbone of a polynucleotide chain is formed by sugar and phosphate.

(ii) In RNA, every nucleotide residue has an additional –OH group present at 3' position in the ribose.

(iii) Uracil is also called demethylated thymine.

(iv) Deoxycytidine is a nucleotides with cytosine as a N-base.

(1) 1 (2) 2 (3) 3 (4) 4

124. Nucleosome :

(A) Formed by wrapping of negatively charged DNA around negatively charged histone octamer.

(B) Typically contain 200 bp of DNA helix.

(C) Constitute the repeating unit of a structure in nucleus called chromatin.

(D) Beads on string.

How many statement are correct.

(1) A, B, C, D

(2) A, B, C only

(3) B, C, D only

(4) C, D only

125. Match column-I with column-II :

	Column-I		Column-II
(a)	Lactose	(i)	Functional part of the gene
(b)	Release factor	(ii)	Inducer of Lac operon
(c)	Histone	(iii)	Packaging of DNA
(d)	Exon	(iv)	Termination of translation

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

126. Choose the incorrect statement :

(1) Goal of HGP was to determine the sequences of the 3 billion chemical base pairs that make up human DNA.

(2) Goal of HGP was to identify the approximately 25000-50000 genes in human DNA.

(3) The HGP was a 13 year project coordinated by the U.S. department of energy and national institute of health.

(4) The sequence of chromosome 1 was completed only in May 2006.

127. Choose the incorrect pair.

(1) Griffith - Transforming principle

(2) Avery, MacLeod and McCarty - Biochemical nature of transforming principle.

(3) Friedrich Meischer - X-ray diffraction.

(4) Watson and Crick - Double helix model of DNA.

128. For a double stranded DNA the ratios between Adenine and Thymine and Guanine and cytosine are constant and equals one, the statements was given by?

(1) Watson and Crick

(2) Erwin Chargaff

(3) Wilkins and Franklin

(4) Messelson and Stahl

129. In dsDNA adenine pair with thymine and guanine pair with cytosine by how many hydrogen bonds respectively?

(1) 1,2 (2) 2,3 (3) 2,1 (4) 3,2

130. The distance between a base pair in a double helix is approximately 0.34 nm and there are roughly 10 bp in each helix, then the pitch of the helix is :-

(1) 34 nm (2) 0.034 nm

(3) 3.4 nm (4) 340 nm

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131. Which of the following event does not take place in prokaryotes-

(1) Synthesis of m-RNA

(2) Initiation of transcription by the help of sigma factor.

(3) Splicing

(4) Translation

132. For initiation of translation the ribosome binds to the mRNA at the start codon (AUG). This codon is recognised only by the :-

(1) sn RNA (2) 23 s rRNA

(3) Initiator tRNA (4) Termination tRNA

133. The y gene of lac operon codes for permease, which increases permeability of the cell to _____.

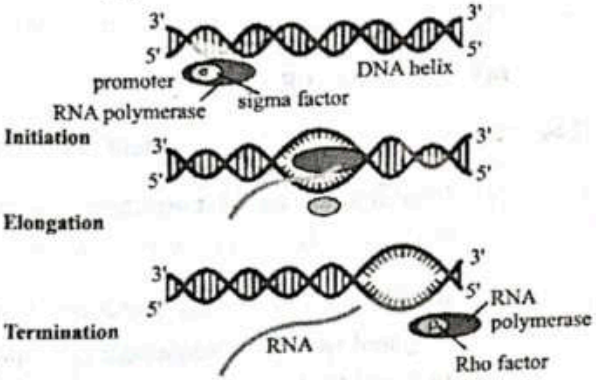
(1) β-galactosides

(2) Glucose

(3) Fructose

(4) Transacetylase

134. Given below diagram represents which of the following process ?



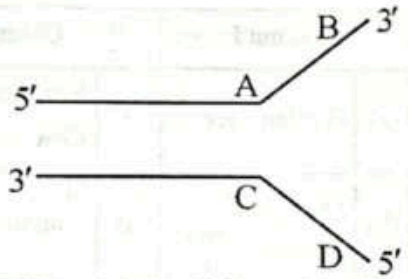
(1) Replication

(2) Transcription in Eukaryotes

(3) Transcription in Prokaryotes

(4) Translation

135. Consider the following replication fork formed by un-winding of DNA molecule



Choose the option that indicates the direction in which replication is continuous.

(1) A → B

(2) B → A

(3) C → D

(4) D → C

136. Transgenic mice are being developed for use in :

(1) Testing the safety of polio vaccines before they are used on human

(2) Molecular diagnosis of diseases

(3) Production of human protein enriched milk

(4) Both (1) and (3)

137. The first transgenic cow 'Rosie', produces :

(1) Human protein-enriched milk (α-lactalbumin)

(2) Human protein α-1 antitrypsin enriched milk

(3) Human protein enriched milk (insulin)

(4) Humulin protein

138. Which of following statement is incorrect about application of biotechnology in agriculture?

(1) Genetically modified crops help in developing herbicide resistant crop.

(2) pBR322 is commonly used as a vector in plants genetic engineering

(3) Golden rice is enriched with vitamin-A

(4) GM crops helped to reduced post harvest losses.

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139. Find out correct match from column I and column II.

Column I		Column II	
(a)	Golden Rice	(i)	First transgenic Cow
(b)	Flavr Savr Tomato	(ii)	Vitamin A
(c)	Rosie	(iii)	Gene Silencing
(d)	RNAi	(iv)	Polygalacturonase

(1) a-ii, b-iii, c-i, d-iv (2) a-ii, b-iv, c-i, d-iii

(3) a-iv, b-i, c-ii, d-iii (4) a-iv, b-ii, c-i, d-iii

140. **Assertion (A) :-** Genetic modification has been used to create tailor-made plants to supply alternative resources to industries in the form of starches, fuels and pharmaceuticals.

Reason (R) :- Plants, bacteria, fungi and animals whose gene have been altered by manipulation are called GMO.

(1) Assertion and Reason are correct but Reason is not the correct explanation of Assertion

(2) Assertion and Reason are correct and Reason is the correct explanation of Assertion

(3) Both Assertion and Reason are incorrect

(4) Assertion is correct but Reason is incorrect

141. **Statement-I :-** Gene therapy is the insertion of genes into an individual's cells and tissues to treat diseases especially hereditary diseases.

Statement-II :- The process of RNA interference has been used in the development of plants resistant to bacteria.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

142. (A) Polymerase chain reaction
(B) Recombinant DNA technology
(C) ELISA test
(D) Urine and serum analysis
How many of these techniques serves the purpose of early diagnosis of a disease for its treatment?

(1) A, B and C only

(2) A and B only

(3) A, B, C & D

(4) B, C & D only

143. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :- The Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Commitee)

Reason (R) :- GEAC makes decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services.

In the light of the above statements, choose the correct answer from the options given below :-

(1) Both A and R are true and R is a correct explanation of A.

(2) Both A and R are true and R is NOT a correct explanation of A.

(3) A is true but R is false.

(4) A is true but R is true.

144. Which of the following statement is incorrect?

(1) Insulin consists of two short polypeptide chains.

(2) In 1997, Eli Lilly an American company prepared two DNA sequences corresponding to A and B chains of human insulin.

(3) ADA enzyme is crucial for the immune system to function

(4) ELISA is based on the principle of antigen antibody interaction

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145. How many statements are correct regarding uses of GM plants.
(A) Made crops more tolerant to abiotic stresses.
(B) Reduced reliance on chemical pesticides.
(C) Helped to reduced post harvest losses.
(D) Decreased efficiency of mineral usage by plants.
(E) Enhanced nutritional value of food.
(1) 2 (2) 3
(3) 1 (4) 4

146. Given below are some steps of formation of insulin by genetic engineering
A. Chain A and B extracted separately
B. Gene of A and B transfer in plasmids of *E. coli* to produce insulin chains
C. Chain A and B combined by creating disulfide bonds
D. Chains A and B were produced separately
E. Prepared two DNA sequences corresponding to A and B chains
Choose the correct sequence of stages/steps from the options given below :-
(1) C-E-D-A-B (2) E-B-D-A-C
(3) B-D-E-A-C (4) E-C-A-D-B

147. As most Bt toxins are insect group specific, so the choice of Bt genes depends upon :-
(1) crop (2) targeted pest
(3) only crop (4) (1) and (2) both

148. The first clinical gene therapy was given to a 4-year old girl with ADA deficiency in :
(1) 1984 (2) 1986
(3) 1992 (4) 1990

149. Which animal is being used to test the safety of polio vaccine?
(1) Transgenic mice (2) Transgenic pig
(3) Transgenic cow (4) Transgenic cat

150. Which of the following statement is incorrect ?
(1) GEAC will make decision regarding the validity of GM research.
(2) There are an estimated 200,000 varieties of rice in India alone.
(3) The Indian Parliament has recently cleared the third amendment of the Indian patents bill.
(4) 27 documented varieties of Basmati rice are grown in India.

151.

	Column-A		Column-B
(A)	Explant	(i)	Nutrient medium for growing plant cell
(B)	Somaclones	(ii)	Part of plant used in tissue culture
(C)	Meristem culture	(iii)	Genetical identical plants
(D)	Sucrose in medium	(iv)	Used to produce virus-free plants

(1) A-ii, B-iii, C-iv, D-i
(2) A-ii, B-iv, C-iii, D-i
(3) A-iii, B-i, C-ii, D-iv
(4) A-iv, B-ii, C-iii, D-i

152.

	Column-I		Column-II
(A)	Cry I Ac	(i)	*E.coli.*
(B)	Cry I Ab	(ii)	*Agrobacterium*
(C)	Ti plasmid	(iii)	Control Cotton boll worms.
(D)	RNA i	(iv)	Control corn bores
		(v)	*Meloidegyne incognitia*

(1) A - (ii), B - (iii), C - (iv), D - (v)
(2) A - (v), B - (iv), C - (iii), D - (ii)
(3) A - (iv), B - (i), C - (ii), D - (iii)
(4) A - (iii), B - (iv), C - (ii), D - (v)

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153. Given below are two statements :

Statement-I : Bt-toxin binds to receptors in the insect's nervous system, paralysing the insect immediately.

Statement-II : Bt-crops need to be sprayed frequently with chemical insecticides for protection.

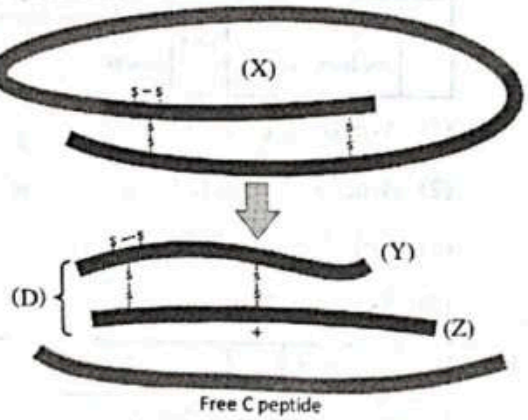
(1) **Statement I** is true but **Statement II** is false.

(2) **Statement I** is false but **Statement II** is true.

(3) Both **Statement I** and **Statement II** are false.

(4) Both **Statement I** and **Statement II** are true.

154. Recognise the figure and choose the correct option given below :



(1) X-Proinsulin, Y-A-Peptide, D-B-Peptide, Z-Mature insulin

(2) X-A-Peptide, Y-Proinsulin, D-B-Peptide, Z-Mature insulin

(3) X-Proinsulin, Y-A-Peptide, D-Mature insulin, Z-B-peptide

(4) None of the above

155. P is a transgenic plant developed by inserting a gene from *Bacillus thuringiensis*. The gene produces a protein that exists as protoxin in bacteria. In the insects gut the protein becomes active and leads to death of insect larvae

Choose correct sequence of events :

(1) W → Conversion of protoxin into active toxin

(2) X → Ingestion by insect (worm)

(3) Y → Insertion of Bt gene in to plant

(4) Z → Expression of cry protein

(1) Y → Z → X → W

(2) Z → Y → X → W

(3) Y → X → Z → W

(4) X → Y → Z → W

156. How many following statements are incorrect ?

(A) Transgenic animals are used in vaccine safety testing.

(B) Rosie, the first transgenic cow, produced human insulin.

(C) Restriction enzyme cleave DNA randomly.

(D) Plasmids are used as vector in rDNA technology.

(1) 4 (2) 3 (3) 2 (4) 1

157. Identify the incorrect statement about Lactic Acid Bacteria (LAB).

(1) They convert milk into curd

(2) They increase nutritional quality of curd

(3) They cause spoilage of milk

(4) They improve gut health

158. Large holes is swiss cheese are due to production of large amount of CO_2 by :-

(1) Bacteria (2) Fungi

(3) Yeast (4) *Lactobacillus*

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English

159. Which of the following is also called brewer's yeast?

(1) *Penicillum notatum*

(2) *Acetobacter aceti*

(3) *Saccharomyces cerevisiae*

(4) *Trichoderma*

160. **Assertion :-** *Trichoderma* is a biocontrol agent of several plant pathogens.

Reason :- Nucleopolyhedro virus are good biocontrol agents for insects.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

161. Which of the following statement is incorrect ?

(1) Lactic acid bacteria plays very beneficial role in checking disease causing microbes.

(2) Wines and beer are produced by fermentation of malted cereals and fruit juices.

(3) Dragon flies are useful bio pesticides which are used to get rid of mosquitoes.

(4) *Azospirillum* and *Azotobacter* are symbiotic bacteria which fixes nitrogen content of soil.

162. Match the column I with column II

Column-I		Column-II	
(a)	*Aspergillus niger*	(i)	Lactic acid
(b)	*Lactobacillus*	(ii)	Butyric acid
(c)	*Clostridium butylicum*	(iii)	Citric acid
(d)	*Trichoderma polysporum*	(iv)	Cyclosporin A

(1) a-iii, b-i, c-ii, d-iv

(2) a-iii, b-ii, c-iv, d-i

(3) a-iii, b-iv, c-i, d-ii

(4) a-iv, b-i, c-ii, d-iii

163. Given below are two statements :

Statement-I : Human insulin is being commercially produced from a transgenic species of *Escherichia coli.*

Statement-II : Human protein, alpha-1-antitrypsin is used to treat emphysema.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

164. Study the following flow chart :

Primary effluent
↓
Aeration tank (with mechanical agitation)
↓
Floc formation
↓
Secondary sedimentation tank
↓
Activated sludge
↓
Anaerobic sludge digester
↓
Biogas production

Which of the following statements are correct?

A. Flocs are aggregates of bacteria associated with fungal filaments.

B. The biogas produced mainly contains methane generated by *Methanobacterium.*

C. Activated sludge is completely discarded and not reused.

D. Aeration promotes growth of aerobic microbes that consume organic matter.

Options :

(1) A, B and D only

(2) A and C only

(3) B and D only

(4) All four statements are correct

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165. Choose the incorrect match.

	Types of microbes	Commercial production	Scientific name
(1)	Fungi	Citric acid	*Aspergilus niger*
(2)	Fungus	Streptokinase	*Monascus purpureus*
(3)	Fungus	Cyclosporin A	*Trichoderma polysporum*
(4)	Yeast	Ethanol	*Saccharomyces cerevisiae*

166. Read the following statements :

(a) Probe is a single stranded DNA or RNA, tagged with a radioactive molecule.

(b) The C-peptide is present in the mature insulin.

(c) RNAi takes place in all eukaryotic organisms as a method of cellular defence.

(d) The protein encoded by the gene cryIAb controls corn borer.

Choose the correct options :

(1) a, b & c (2) b, c & d

(3) a, b & d (4) a, c & d

167. Read column-I, II & III and identify their correct match :

	Column-I		Column-II		Column-III
(A)	*Bacillus thuringiensis*	(P)	Restriction endonuclease	(i)	Cloning of foreing DNA
(B)	Rosie	(Q)	Insecticidal protein	(ii)	Produce sticky ends
(C)	Plasmid	(R)	Transgenic cow	(iii)	Kill certain insect
(D)	EcoRI	(S)	Cloning vector	(iv)	Human protein enriched milk

(1) A-(Q,iii); B-(R,iv); C-(S,i); D-(P,ii)

(2) A-(R,iv); B-(Q,iii); C-(P,ii); D-(S,i)

(3) A-(Q,iv); B-(R,iii); C-(P,i); D-(P,i)

(4) A-(S,i); B-(P,ii); C-(Q,iii); D-(R,iv)

168. Read the following statements :-

(a) Over 95 percent of all existing transgenic animals are mice.

(b) The Indian government has set up organisation such as GEAC, which will make decisions regarding the validity of GM research.

(c) If any protein encoding gene is expressed in a homologous host, it is called as recombinant protein.

(d) The most commonly used bioreactors are of stirring type.

(e) The downstream processing and quality control testing vary from product to product.

Choose the correct options :

(1) a, b, c & d

(2) a, b, d & e

(3) b, c, d & e

(4) a, b, c, d & e

169. **Statement-I** :- Recombinant DNA technology, PCR and ELISA are some of the techniques that serve the purpose of early diagnosis.

Statement-II :- PCR is now routinely used to detect HIV in suspected AIDS patients.

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

170. **Statement-I** :- *Bacillus thuringiensis* produce proteins that kill coleopterans and dipterans.

Statement-II :- Bt toxin requires alkaline pH for activation.

(1) Statement I is correct, statement II is incorrect

(2) Statement I is incorrect, statement II is correct

(3) Both statement I & statement II are incorrect

(4) Both statement I and statement II are correct

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171. Which of the following triggers protection against nematode infestation.

(1) Host plant generated dsRNA

(2) Host plant generated ssRNA

(3) Nematode generated dsRNA

(4) Nematode generated ssRNA

172. The capacity to generate a whole plant from any cell/explant is called _____.

(1) Somatic hybridisation

(2) Totipotency

(3) Tissue culture

(4) Micropropagation

173. **Statement-I** :- Isolated protoplasts from two different varieties of plants – each having a desirable character can be fused to get hybrid protoplasts, which can be further grown to form a new plant. These hybrids are called *Somaclones.*

Statement-II :- By application of tissue culture it is possible to achieve propagation of a large number of plants in very short duration. This method of producing thousands of plants through tissue culture is called somatic hybridisation.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

174. Match Column-I with Column-II for regulatory and ethical aspects in biotechnology:

Column-I		Column-II	
(A)	GEAC	(i)	Legal right for invention or process
(B)	Patent	(ii)	Unauthorized use of biological resources
(C)	Biopiracy	(iii)	Approval of GM research and release in India.
(D)	Bioethics	(iv)	Moral concerns in the use of biotechnology

(1) A-iii, B-ii, C-i, D-iv

(2) A-ii, B-iii, C-iv, D-i

(3) A-iv, B-ii, C-iii, D-i

(4) A-iii, B-i, C-ii, D-iv

175. RNAi takes place in all _____ organisms as a method of cellular ____

(1) prokaryotic, attack

(2) eukaryotic, attack

(3) prokaryotic, defense

(4) eukaryotic, defense

176. How many recombinant therapeutics have been approved at present for human use the world over ?

(1) 30 (2) 12 (3) 27 (4) 2

177. During gene therapy of adenosine deaminase (ADA) deficiency a functional ADA cDNA is introduced into lymphocytes by using a :

(1) plasmid of *E.coli* (2) phage DNA

(3) retroviral vector (4) Ti plasmid

178. Yeast is used in the production of :-

(1) Cheese and curd

(2) Bread and beer

(3) Citric acid and Cyclosporin A

(4) Cheese and acetic acid

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179. Select incorrect statement :-

(1) Dung can be used for generation of biogas, commonly called gobar gas

(2) Both *Azotobacter* and *Rhizobium* are free living in the soil and can fix atmospheric nitrogen in soil

(3) Baculoviruses are pathogens that attack insects and other arthropods

(4) Indirectly BOD is a measure of the organic matter present in the water

180. Which of the following acts as competitive inhibitor for the enzyme responsible for synthesis of cholesterol ?

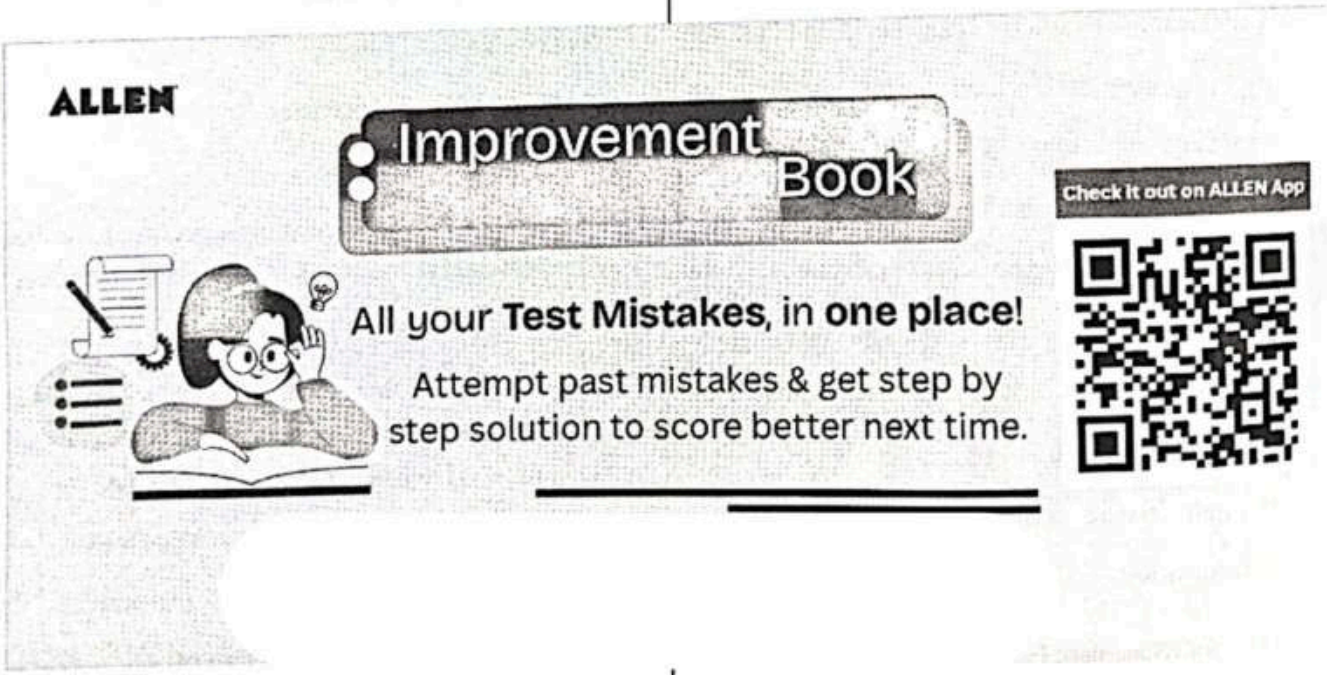
(1) Cyclosporin-A

(2) Statins

(3) Streptokinin

(4) Citric acid





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